

Q02 - EDA

Sample mean $\rightarrow m(x) = \frac{1}{N} \sum_{i=1}^N x_i$

Sample covariance $\rightarrow \text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N ((x_i - m(x))(y_i - m(y)))$

Sample variance $\rightarrow s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(x))^2$

(1) $m(a+bX) \rightarrow \frac{1}{N} \sum_{i=1}^N (a+bX)$
 $\rightarrow a + \left(\frac{1}{N} \sum_{i=1}^N (bX) \right) \rightarrow a + b \left(\frac{1}{N} \sum_{i=1}^N (x) \right)$
 $\rightarrow \boxed{a + b(m(x))}$

(2) $\text{cov}(X, a+bY) \rightarrow \frac{1}{N} \sum_{i=1}^N ((x - m(x))(a+bY - m(a+bY)))$
 $\rightarrow \frac{1}{N} \sum_{i=1}^N ((x - m(x))(a+bY - (a+b(m(Y))))$
 $\rightarrow \frac{1}{N} \sum_{i=1}^N ((x - m(x))(b(Y - m(Y))))$
 $\rightarrow b \left(\frac{1}{N} \sum_{i=1}^N ((x - m(x))(Y - m(Y))) \right)$
 $\rightarrow \boxed{b \times \text{cov}(X, Y)}$

(3) $\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N ((a+bX - m(a+bX))(a+bX - m(a+bX)))$
 $\rightarrow \frac{1}{N} \sum_{i=1}^N (a+bX - (a+b(m(X)))(a+bX - (a+b(m(X))))$
 $\rightarrow \frac{1}{N} \sum_{i=1}^N (b(X - m(X)))(b(X - m(X)))$
 $\rightarrow b^2 \left(\frac{1}{N} \sum_{i=1}^N (X - m(X))(X - m(X)) \right)$
 $\rightarrow \boxed{b^2 \times \text{cov}(X, X)}$

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$$(3) \text{cov}(X, X) \Rightarrow \frac{1}{N} \sum_{i=1}^N (X - m(X))(X - m(X))$$

$$\hookrightarrow \boxed{\frac{1}{N} \sum_{i=1}^N (X - m(X))^2 = S^2}$$

$$(4) g() = \text{non-decreasing}$$

$$\text{Ex: } X = [1, 2, 3] \quad g() = \log()$$

$$m(g(X)) \rightarrow m(\log(X)) \rightarrow \frac{1}{3} \sum_{i=1}^3 \log(x)$$

$$\rightarrow (0 + 0.301 + 0.477) / 3 \approx \boxed{0.259}$$

$$g(m(X)) \rightarrow g\left(\frac{1+2+3}{3}\right) \rightarrow g(2) \rightarrow \log(2)$$

$$\approx 0.301$$

$$\boxed{0.259 \neq 0.301}$$

$\therefore m(g(X)) = g(m(X))$ is NOT always true